

Simple Linear Regression

Thursday, September 24 & Monday, September 28, 2026

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Quantitative Reasoning II · INF 1220 · University of Austin

Regression

Regression is the most widely used statistical tool for understanding relationships among variables. It writes the relationship as an equation connecting

- an outcome (**dependent variable**)
- to one or more factors (**independent variables**, or **predictors**)

Why?

Straight up **Prediction:**

- What will I sell my house for?

Explanation and understanding:

- Does building an addition raise the value of my house?
- How does my diet affect my health?
- What is the impact of economic freedom on economic growth?

Predicting House Prices

Problem: Predict market price based on observed characteristics

Solution:

- Look at property sales data where we know both the price and some observed characteristics
- Build a model that predicts price as a function of observed characteristics

Which characteristics?

Defining the Variables

To keep things super simple, let's focus on size only.

Y = sale price of the house, in dollars

X = above-grade living area, in square feet

Y is the **dependent variable** (outcome): the variable we are trying to predict.

X is the **independent variable** (input): the variable we use to predict.

Where This Data Comes From

2,930 residential sales in **Ames, Iowa**, 2006 to 2010.

A professor at Truman State published this dataset in 2011 specifically for statistics education. It contains 82 variables with clear definitions, and it represents real sales in

Ames, Iowa.

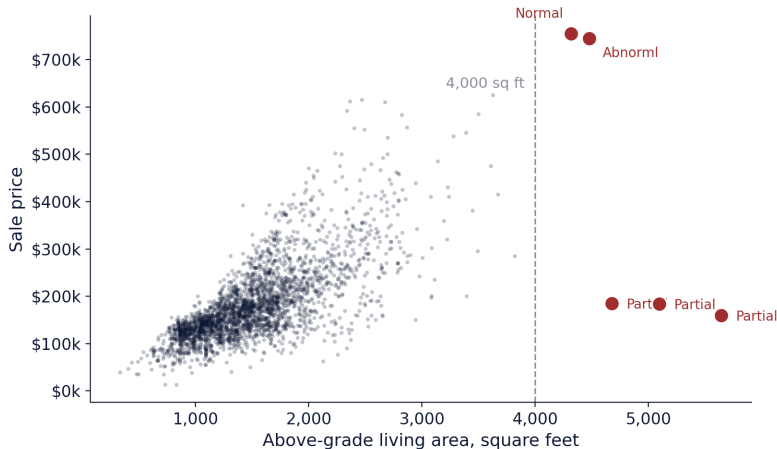
Before Anything Else

The dataset's manual tells us to throw some sales out, and why:

"There are five observations that an instructor may wish to remove. . . Three of them are true outliers (Partial Sales that likely don't represent actual market values) and two of them are simply unusual sales (very large houses priced relatively appropriately). I would recommend removing any houses with more than 4000 square feet."

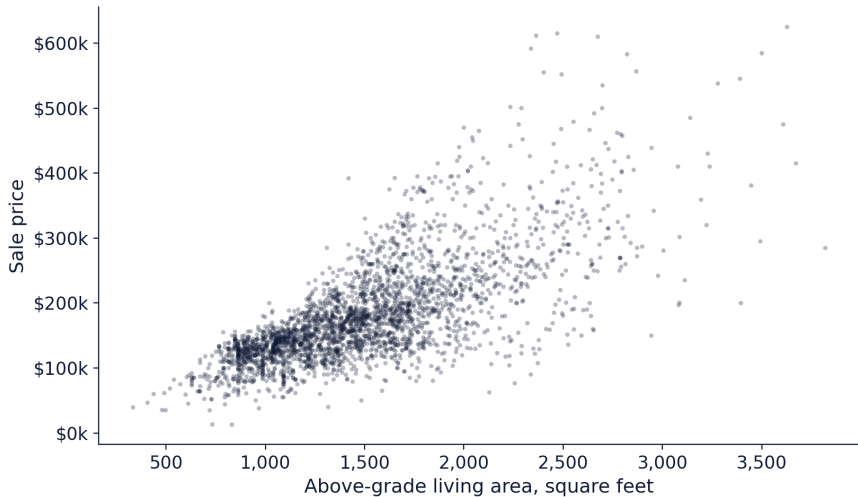
That leaves **2,925** sales for us to work with today

A visual of the dropped sales



There are many ways to deal with outliers. When creating analyses, it's important to document the choices you make.

Scatter Plot: Price vs. Size



Correlation

Correlation is a measure of how related two variables are.

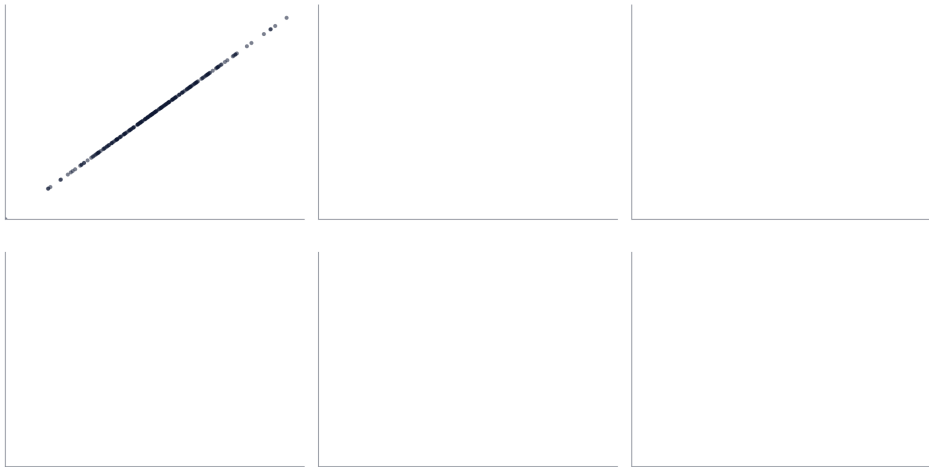
$$-1 \leq \text{corr}(X, Y) \leq 1$$

A correlation of 1 means the variables move together perfectly in a straight line.

$\text{corr}(X, Y) = \text{corr}(Y, X)$ because correlation is symmetric

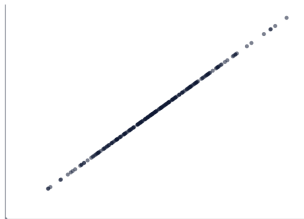
Correlation

correlation = +1.00

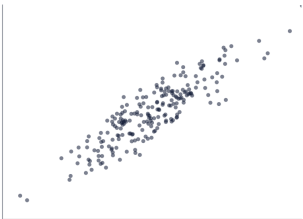


Correlation

correlation = +1.00

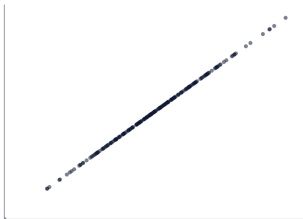


correlation = +0.90

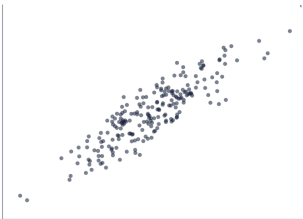


Correlation

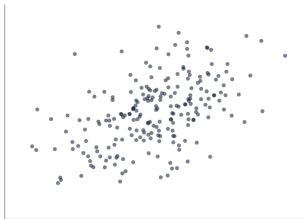
correlation = +1.00



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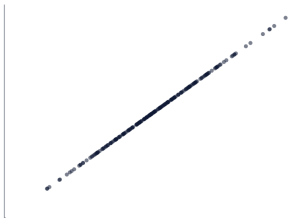


correlation = +0.50

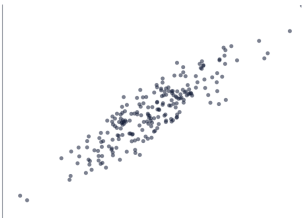


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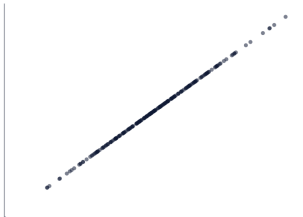


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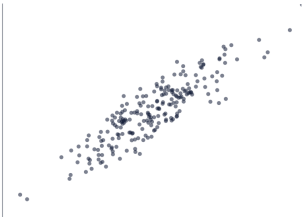


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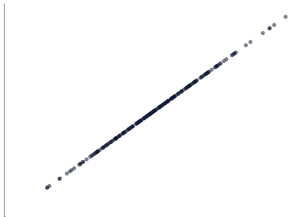


correlation = -0.50

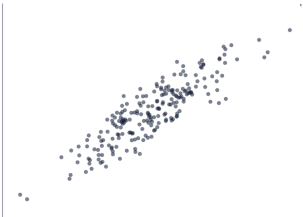


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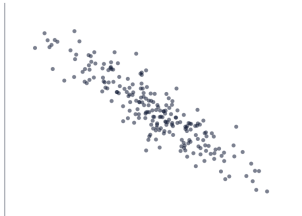
correlation = +0.00



correlation = -0.50

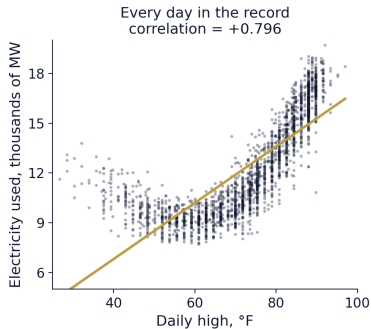


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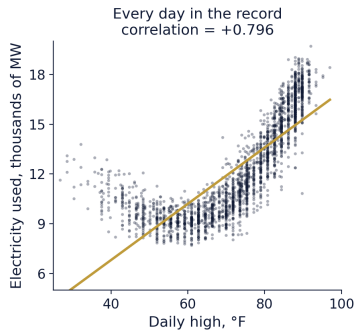
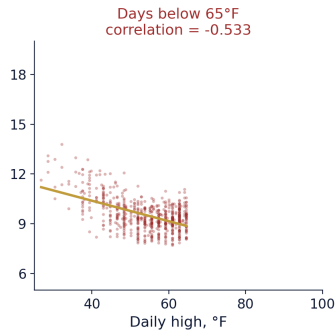
Nonlinear relationships

Daily electricity use in Houston against the daily high temperature. **2,357** days.



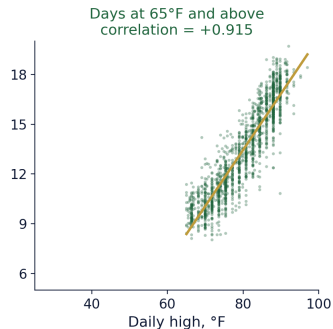
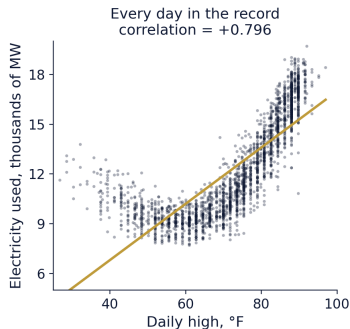
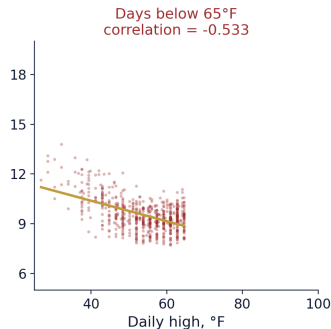
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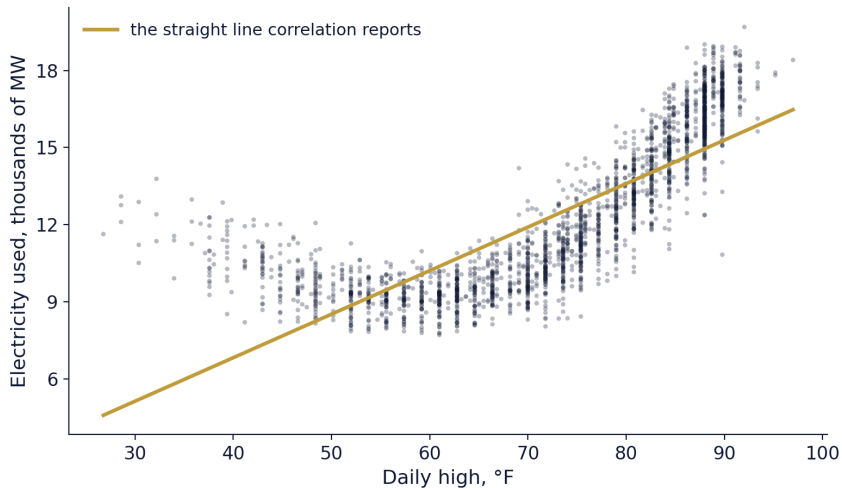
The Correlation Changed Sign

On cold days, warmer day $\rightarrow$ people stop heating their homes $\rightarrow$ less electricity used.
 $\text{corr} = -0.533$.

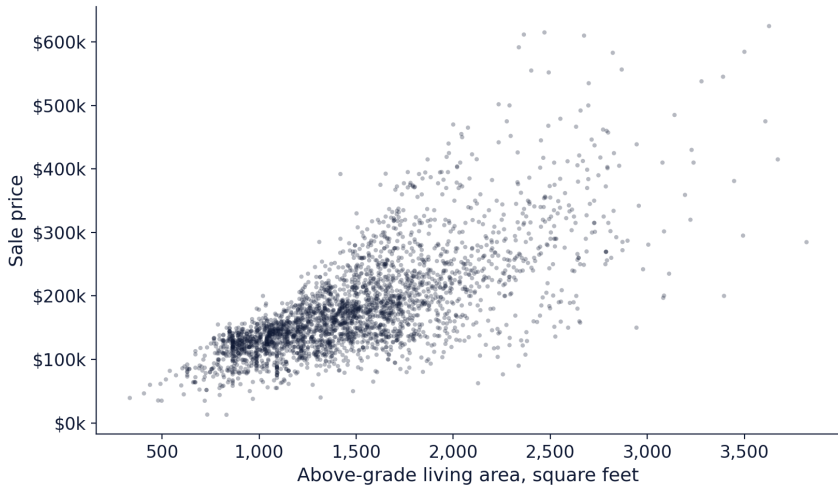
On warm days, warmer day $\rightarrow$ people turn on AC $\rightarrow$ more electricity used.
 $\text{corr} = +0.915$.

Over the whole record, $\text{corr} = +0.796$ - why?

Daily Electricity Use vs. Temperature - Houston, TX



Sale Price vs. Living Area - Ames, IA



Sale Price vs. Living Area - Ames, IA

$$\text{corr}(\text{area}, \text{price}) = +0.719$$

Positive, strong, and reasonably straight

Fitting a straight line is a reasonable thing to do here

Always check the scatterplot before fitting a line – just because a correlation is strong doesn't mean a straight line is a good fit.

What does our fitted line tell us?

Not: what price will this particular house sell for?

But: **what is the average price of homes of this size, in our data?**

A fitted line is the average Y at each X in our data.

Simple Linear Regression Model

The line we fit is a model of the relationship between price and size, represented mathematically as:

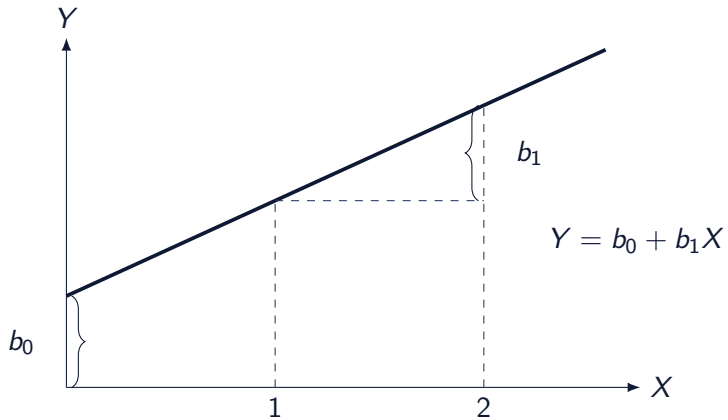
$$Y = b_0 + b_1X + e$$

b_0 is the **intercept**, in units of Y : dollars.

b_1 is the **slope**, in units of Y per unit of X : dollars per square foot.

e is the **error** or **residual**: the difference between the actual price and the predicted price.

Simple Linear Regression Model



How Do We Find the Best Line?

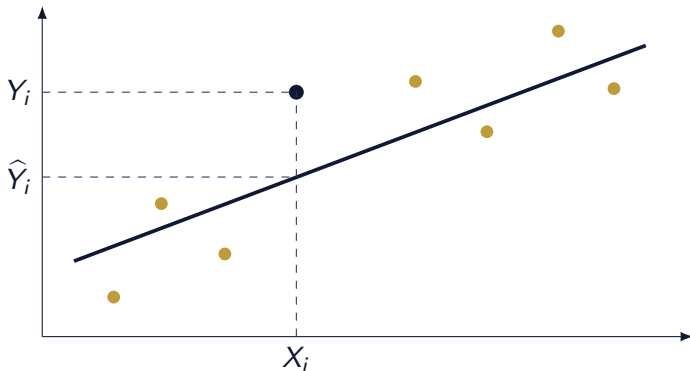
We need a rule for choosing b_0 and b_1 in $\hat{Y} = b_0 + b_1X$.

A reasonable way to fit a line is to minimise the amount by which the fitted value differs from the actual value.

In other words, we want to minimise the **error** or **residual**.

Fitted Values vs Residuals

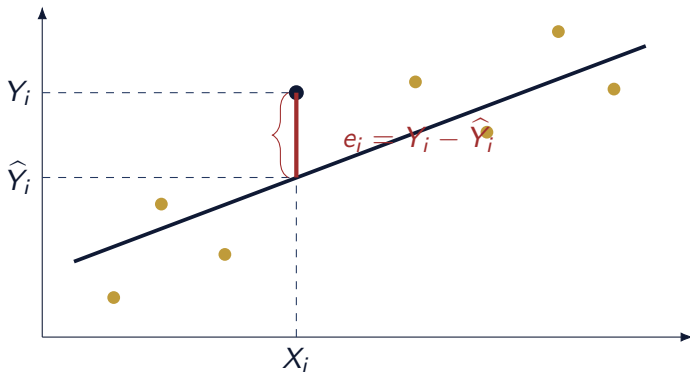
What is the fitted value?



The dots are the observed values. The line is the fitted values, $\hat{Y}_i = b_0 + b_1 X_i$.

Fitted Values vs Residuals

What is the residual for the i th observation?



$$Y_i = \hat{Y}_i + e_i$$

Least Squares

Ideally we want every residual to be zero.

- If they were, we would have a perfect line.
- Moving closer to some points moves you away from others.

Why do we square the residuals?

- Ensures that positive and negative residuals don't cancel each other out

Least Squares

Ideally we want every residual to be zero.

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The fitting process:

- Choose b_0 and b_1 to minimise the sum of squared residuals (SSE)
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Why do we square the residuals?

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Our Regression Model

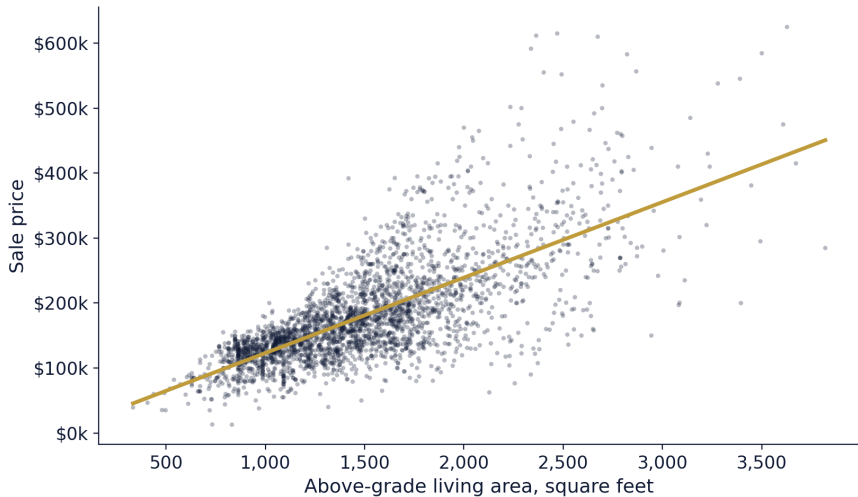
$$\widehat{\text{price}} = 6,773 + 116.23 \times \text{square feet}$$

Slope: \$116.23 per additional square foot

Intercept: \$6,773. The fitted price of a house with no square footage

In this model, the intercept represents a house that doesn't exist

Line of Best Fit

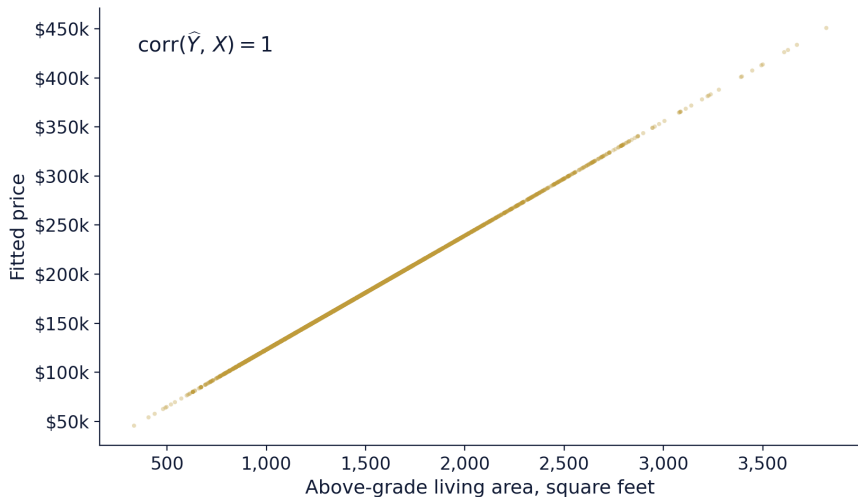


More on Least Squares

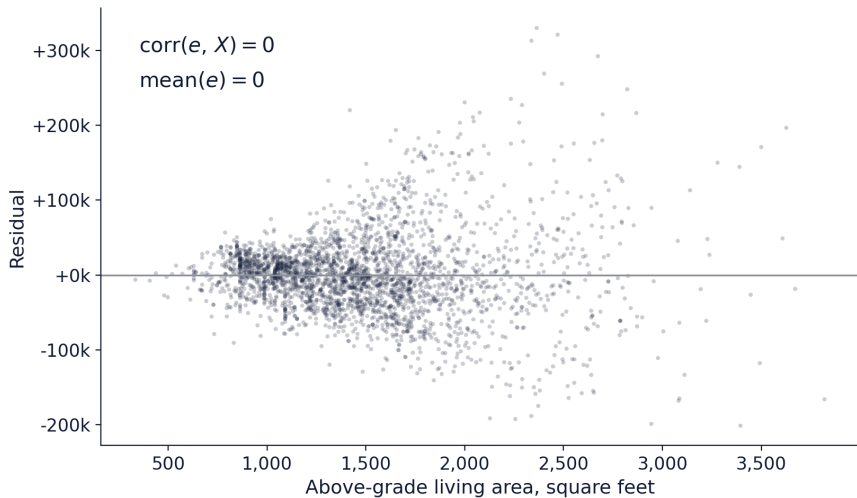
From here on, fitted values $\hat{Y}_i$ and residuals e_i mean the least-squares ones.

Those fitted values and residuals have two properties. The next two frames show them on these 2,925 sales.

The Fitted Values and X are perfectly correlated

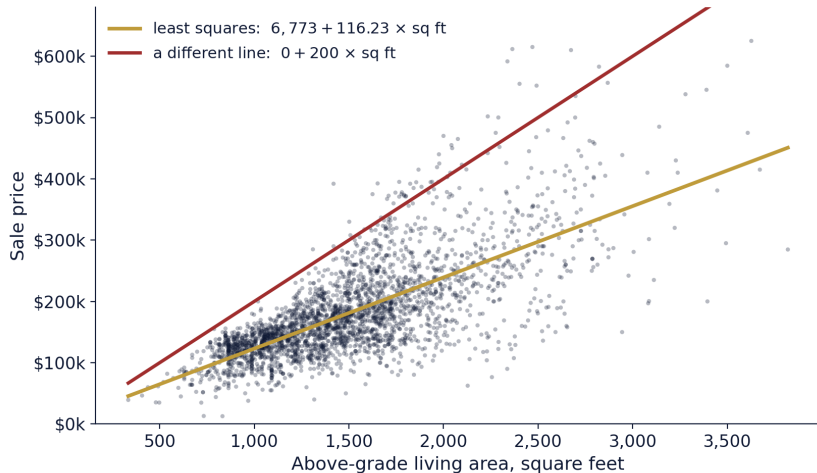


The Residuals and X are uncorrelated



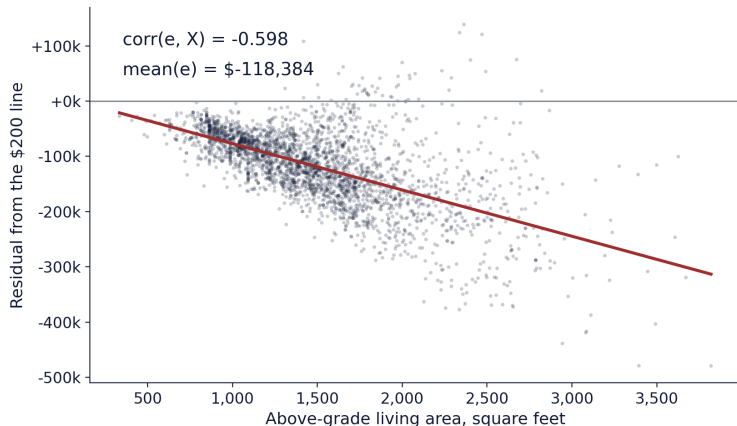
Why Those Two Properties

Let's say we chose a different slope: $\widehat{\text{price}} = 200 \times \text{square feet}$.



X and the residuals are correlated

This line overestimates



This is a way to test our model: if the residuals and X are correlated, a different slope would be a better fit.

Some of the special properties of $\hat{Y}$ and e

$$Y = \hat{Y} + e$$

$\hat{Y}$ is made from X : $\text{corr}(X, \hat{Y}) = 1$.

e is unrelated to X : $\text{corr}(X, e) = 0$.

R^2 - Goodness of Fit

R^2

- is a measure of how well our model fits the data
- tells us how much of the variation in Y our model explains
- is in the range 0 to 1

$$R^2 = 0.518 \text{ for our model}$$

interpretation: size explains about half the variation in sale price

A higher R^2 does not make a model correct

Exercise: Comparing and Interpreting Models